Kavish Wadehra

+16472135948 — kavishwadehra@gmail.com — Linkedin — Personal Portfolio

EDUCATION

Master of Applied Science – Electrical Engineering May 2026 – May 2028
McMaster University, Hamilton, ON
GPA: 4.00 / 4.00 (Dean's List)

Bachelor of Engineering – Electrical Engineering Sep 2020 – Apr 2026
McMaster University, Hamilton, ON
GPA: 3.8 / 4.00 (Dean's List)

SKILLS

Power Electronics & HV Systems: DC-DC converters, Battery Disconnect Unit (BDU) design, pre-charge circuits, battery pack architecture, EVT/HVT testing, power validation, failure analysis
Embedded Systems & Firmware: C, C++, Python, MATLAB, shell scripting, ESP32 systems, PIC18F microcontrollers, SPI, I2C, CAN, LIN, UART
Programming & Data: Python (NumPy, SciPy, TensorFlow, NI-DAQmx API), object-oriented programming (C++/Java), basic machine learning techniques, data acquisition and analysis
Hardware & PCB Design: Altium Designer, PCB schematic and layout, board bring-up and debugging, ADCs, sensor integration, soldering, circuit design, oscilloscopes
Modeling, Simulation & CAD: MATLAB/Simulink, LTspice, NI Multisim, LabVIEW, SolidWorks, Autodesk Inventor, Fusion 360, Eagle, Ansys Granta
Battery Systems: Orion 2 BMS, NXP BMS hardware, voltage/current/temperature sensing, system integration
Tools & Platforms: Git, Linux/Unix, Windows, JIRA, Confluence, Microsoft Office
Soft Skills: Technical communication, cross-functional collaboration, organization, independent problem-solving, team leadership

EXPERIENCE

Technical Program Manager Intern – Vehicle Power Electronics
Tesla, Palo Alto, CA May 2025 – Sep 2025

- Led cross-functional execution of vehicle power electronics programs, coordinating hardware design, validation, test, and manufacturing teams to maintain alignment across program milestones.
- Supported Functional Validation Testing (FVT), Hi-Pot testing, and New Product Introduction (NPI) builds for high-power electronic assemblies, reviewing test results and identifying failure trends.
- Tracked technical and manufacturing blockers, working with engineering teams to define mitigation plans and escalate risks during bi-weekly executive readiness reviews.

Place-and-Route (PnR) CAD Co-op
Advanced Micro Devices (AMD), Markham, ON May 2023 – Apr 2024

- Investigated and helped resolve 60+ issues across PnR and VLSI flows, including QoR degradation, runtime regressions, tool crashes, and flow instability.
- Used ICC2, Fusion Compiler, and Innovus to inspect physical layouts and debug timing, congestion, and power-related design issues.
- Developed automation tools in Python, shell scripting, and C++ to track team-wide performance metrics, disk usage, and testcase packaging.
- Worked in a Linux environment to create cron jobs and automated tasks, reducing manual intervention in daily CAD workflows.

Hardware Manager
Battery Workforce Challenge, Hamilton, ON May 2024 – May 2026

- Led a 30-member cross-functional team developing automotive battery pack electronics, coordinating electrical, mechanical, and software efforts with faculty advisors.
- Designed, brought up, and tested multiple PCBs in Altium, including ESP32-based temperature acquisition boards, flex PCBs for voltage taps, custom cell monitoring units, various boards for monitoring VIT data.

- Programmed ESP32 firmware in C for SPI-controlled ADCs and CAN bus communication; validated designs using oscilloscopes, multimeters, and bench testing.
- Created system-level schematics and mechanical integration designs in SolidWorks for battery disconnect units, busbars, mounting hardware, and module-level enhancements.

Electrical Team Manager
McMaster Solar Car Project, Hamilton, ON — Oct 2020 – Sep 2024

- Led a 40-member electrical team responsible for the design and integration of the university's first functional multi-occupant solar vehicle.
- Performed lithium-ion cell characterization and configured the Orion BMS for vehicle-level battery management and safety monitoring.
- Designed, tested, and assembled multiple PCB revisions in Altium, including:
  - Power Management Board (4 revisions) controlling contactors using PIC18F and ESP32 with SPI and CAN communication.
  - Pedal Control Board (2 revisions) implementing throttle input, regenerative braking, and cruise control.
  - Pre-Charge Board (3 revisions) enabling controlled contactor sequencing and inrush current management.
- Supported vehicle integration, system testing, and electrical troubleshooting across subsystems.

Lab Assistant
Center for Mechatronics and Hybrid Technologies (CMHT), Hamilton, ON — May 2022 – Apr 2023

- Developed LabVIEW and Python scripts for automated data acquisition using NI-DAQmx drivers across multiple experimental setups.
- Performed electrical signal processing and analysis on Bi-Polar Operational Amplifier circuits and sensor interfaces.
- Collected and analyzed experimental data from various laboratory stations, motors, and electromechanical systems.
- Conducted electrical testing and characterization of 18650 lithium-ion cells using signal acquisition and processing techniques.
- Supported laboratory setup, troubleshooting, and validation to ensure repeatable and reliable experimental results.

Undergraduate Teaching Assistant
McMaster University, Hamilton, ON — Jan 2022 – Present

- Served as a Teaching Assistant for Physics 1E03 (Winter 2022, Winter 2026) and Engineering Physics 1P13A (Fall 2022).
- Assisted students with problem-solving in mechanics, electricity and magnetism, and introductory engineering physics.
- Supported laboratory sessions and tutorials, guiding students through experimental procedures and data analysis.
- Evaluated exams, laboratory reports, and design studio submissions, providing technical feedback and grading support.
- Communicated complex technical concepts clearly to students with diverse academic backgrounds.

PUBLICATIONS

Wadehra, K., et al. *Thermal Performance Evaluation and Design Optimization of a Battery Disconnect Unit for High-Power EV Applications.* IEEE ITEC/EATS, 2025.